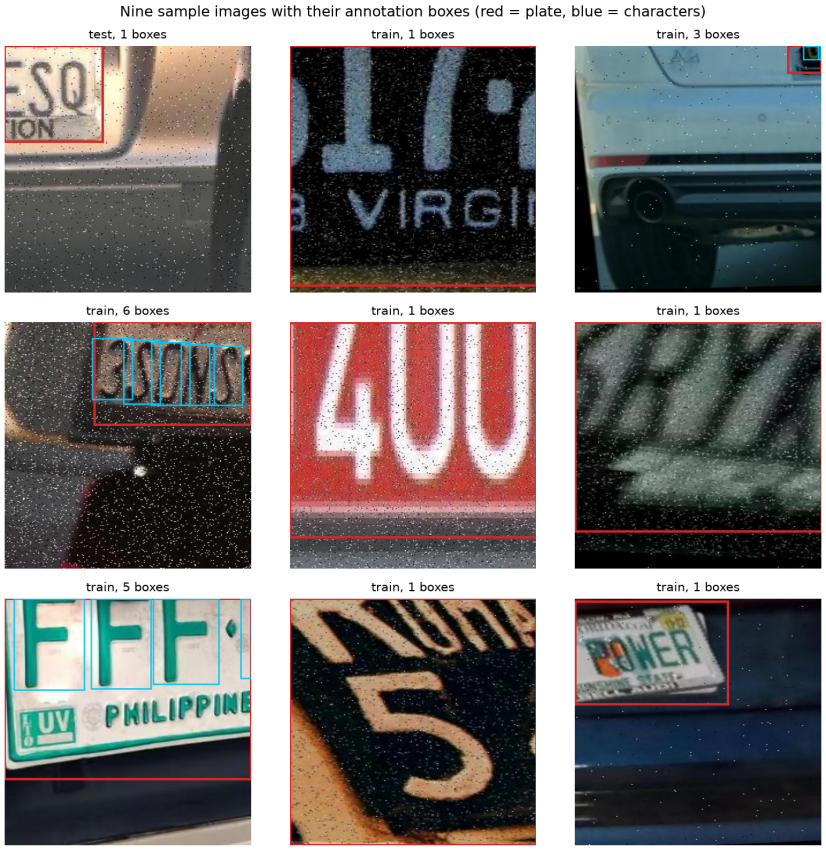
Pakistan car number plate dataset: a look at the annotations

Kaggle: musakhanalabazy/pakistan-car-number-plate and musakhanalabazy/pakistan-car-number-plate-data-set (identical contents). Roboflow export, YOLO format.

What this dataset is

This is an object-detection dataset rather than a table: 12,322 images, all 640 by 640 pixels, each with a text file listing the boxes drawn on it. There are 37 classes, the ten digits, the 26 letters and one class called LicensePlate. The author published it under two Kaggle listings whose archives are identical, so they are treated as one project here. The files are split into 11,172 training, 664 validation and 486 test images.

To analyse it I read every label file into one table with a row per box (24,534 rows including a placeholder row for each image that has no labels), and worked from that table. The images themselves were only opened to check their size, to fingerprint the files, and to draw the small sample sheet below. So this report is about what is annotated and how the files are organised. It does not judge whether the labels are correct.



Nine labelled images chosen at random (fixed seed). Red is the plate box, blue the character boxes. Most are tight crops with speckle noise, and several plates are plainly not Pakistani.

What was checked and cleaned

The label files are not uniform. 19,014 lines are ordinary boxes, but 4,297 lines are polygons with between four and sixteen corner points, and all of those belong to the LicensePlate class. That is 37.3% of the plate annotations. Code that expects five numbers per line would misread them, so each polygon was replaced by the rectangle that encloses it.

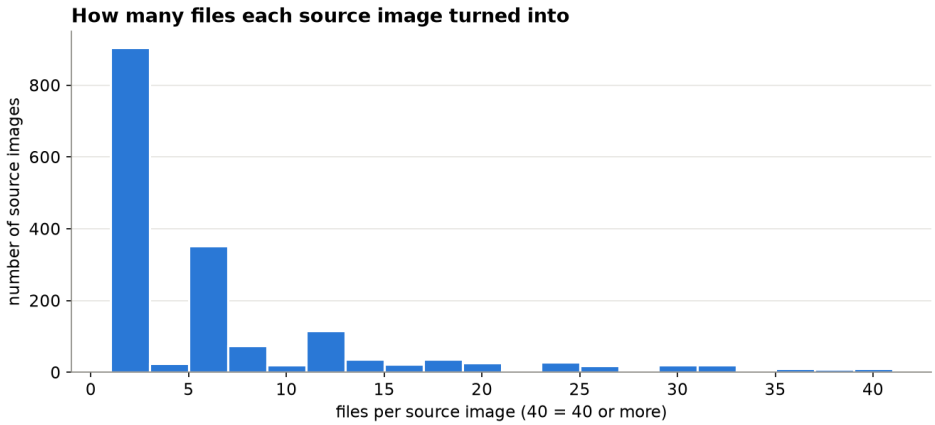
Beyond that the labels are tidy. Every image has a label file, all class ids are valid, and no box reaches outside its image. Thirteen plate boxes were less than two pixels across in one direction, the remains of a plate that a crop almost entirely removed, and these were dropped. Forty-nine image files turned out to be byte-for-byte copies of another file. Twelve of those pairs sit in two different splits, and in four pairs the two copies are labelled differently (one has only the plate, the other has the characters too). I kept the copy with the fuller annotation and removed the other.

The 1,223 images with an empty label file were kept and flagged rather than removed. After cleaning there are 12,273 images, 11,465 plate boxes and 11,793 character boxes; the cleaned table has 24,482 rows against 24,534 before, with extra columns for the source image, the number of copies of that source, and whether the source appears in more than one split.

The main finding: far fewer photos than files

File names follow the Roboflow pattern of a source name, then ".rf.", then a hash. Stripping the hash leaves only 1,748 distinct source images behind the 12,322 files. On average each source appears about seven times, the median is two, and the most repeated one appears 54 times. Every source has an even number of copies.

The Roboflow readme shipped with the data states that no augmentation was applied. The copies are nevertheless different files: different crops or tiles of the same photo, with speckle noise and brightness changes, as the sample sheet shows. The likeliest reading is that the augmented tiles were made earlier and uploaded as if they were originals, so the readme is accurate only about the last export step. The files do not record which transformations were used.



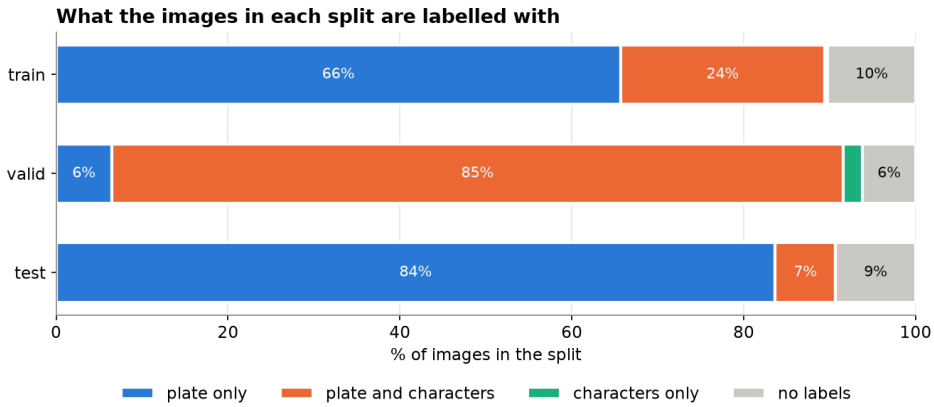
Number of files produced from each of the 1,748 source images.

This matters because the train, validation and test split was made per file, not per source. 249 sources (14.2%) have copies in more than one split and 32 are in all three. Counted by image, 352 of the 486 test images (72.4%) and 284 of the 664 validation images (42.8%) come from a photo that is also in the training set. A model scored on the shipped test set is therefore being tested largely on photos it has already seen in another crop, and the score will flatter it.

What is actually labelled

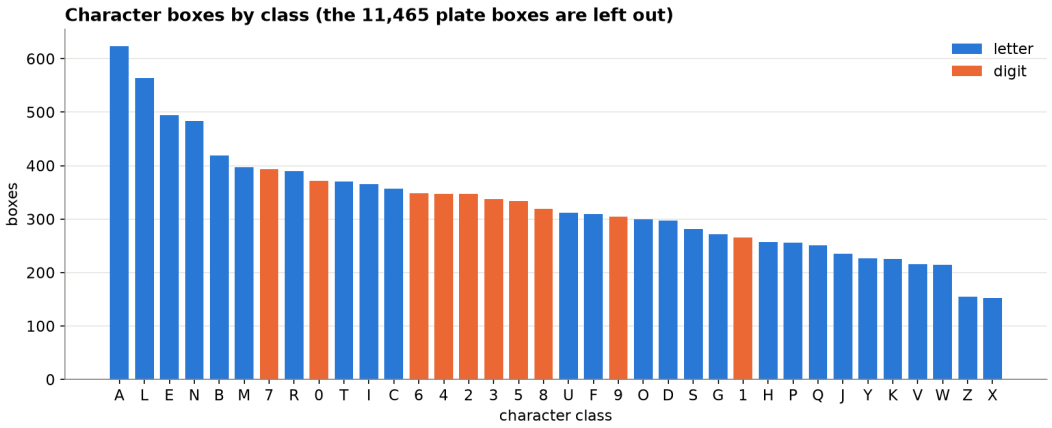
With 36 character classes one would expect a plate-reading dataset, but it is mostly a plate-finding one. 63.2% of the images carry only a plate box, 26.4% have a plate and characters, 0.4% (54 images) have characters with no plate box, and 10.0% have nothing labelled. Most of the unlabelled images, 1,130 of 1,224, come from a source whose other tiles do have a plate, which suggests they are background tiles and not forgotten annotations. It is not the case that every image has exactly one plate: 378 have two or more, and three have nine.

The three splits are also quite different from each other. Character boxes appear in 24.1% of training images, 87.3% of validation images and only 7.0% of test images. The whole test split contains 101 character boxes; every character class has fewer than ten examples there and some have none. The test set can say something about plate detection but next to nothing about character recognition.



Share of images in each split by what is labelled on them.

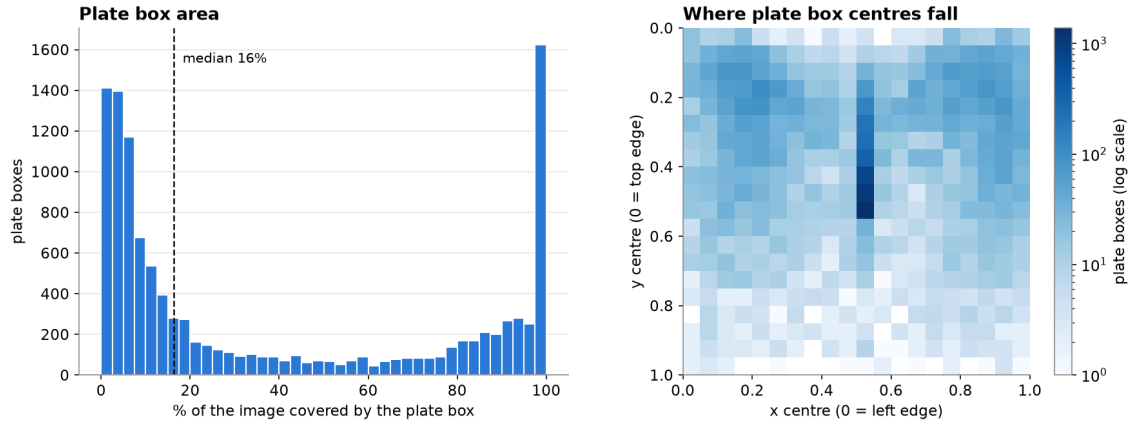
Among the characters themselves the balance is reasonable. A is the most common with 624 boxes and X the rarest with 152, a ratio of about four to one, and the digits range from 393 (the digit 7) down to 266 (the digit 1). Letters outnumber digits, 8,425 boxes to 3,368. These counts include repeated crops, though. Counted once per source photo, Z appears in just 60 photos and X in 75, and only 1,140 of the 1,748 sources have any character labelled.



Character boxes per class after cleaning.

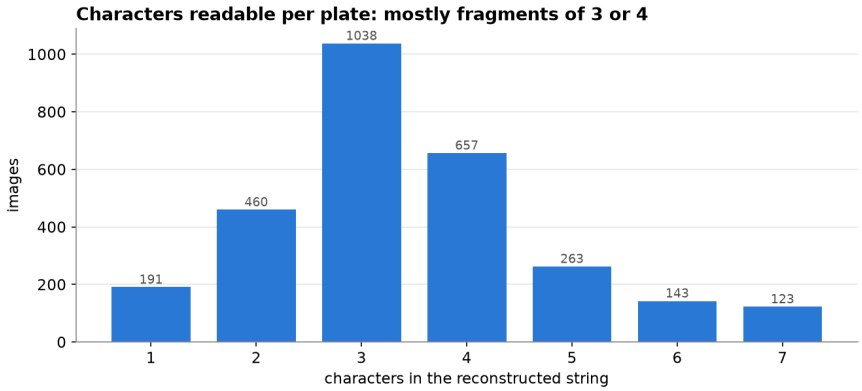
Box sizes, and why whole plates are rare

The plate boxes are large. The median one measures 338 by 218 pixels and covers 16.3% of the image, 35.9% of them cover more than half the frame, and 91.8% touch an image edge. Only 4.6% cover under one percent of the image, the situation you would expect from a photo of a whole car. About one plate box in six (16.1%) is taller than it is wide, which comes from the stretch to a square image and from crops that slice a plate.



Left: share of the image covered by each plate box. Right: where plate box centres fall; the dark column in the middle is the boxes spanning the full image width.

I also tried reading the plate text by ordering the character boxes from left to right inside the plate box. This was limited to images with exactly one plate whose characters sit on a single line, which gave 2,875 strings from 998 source photos. It is an approximate exercise and no strings were printed or stored. The result is mainly a negative one: the median string has three characters, only 18.4% have five or more, and the longest has seven. 79.3% start with a letter, and just 6.5% have the two-or-three letters followed by two-to-four digits shape of a standard Pakistani plate. The images mostly show fragments of plates, so the dataset cannot be used to study plate numbers or registration prefixes.



Length of the character string that could be reconstructed per image (approximate).

Is it really Pakistani?

Only partly, as far as file names and a few sampled images can tell. 284 sources are named like car-sale listings with a model, year and listing number, and those look local. But 94 sources are named "syn_" plus a number, 412 carry "augmented" in their original name, five are named after an image generator, and others contain Dutch and Spanish words for Spain, the Netherlands and number plates. The sample sheet shows American and Philippine plates. I could not measure the Pakistani share from annotations, so the honest description is a mixed-country plate collection under a Pakistani title.

Highest, lowest, most

- 12,322 files come from 1,748 source images; the most repeated source has 54 files.
- 249 sources (14.2%) appear in more than one split; 32 appear in all three.
- 72.4% of test images and 42.8% of validation images share a source photo with the training set.
- LicensePlate is the largest class with 11,465 boxes, 49.3% of all boxes.

- Most common character: A with 624 boxes. Rarest: X with 152. Most common digit: 7 (393); rarest: 1 (266).
- The test split holds only 101 character boxes, against 9,681 in train and 2,011 in validation.
- 1,224 images (10.0%) have no labels; 3,296 (26.9%) have at least one character box.
- The busiest image has 17 boxes; the most plate boxes in a single image is nine.
- 4,297 plate annotations (37.3%) were stored as polygons instead of boxes.
- 91.8% of plate boxes touch an image edge; the median plate box is 338 by 218 pixels.
- Image files range from 7.6 KB to 136 KB, with a median of 39 KB.

Conclusion and limits

The dataset is usable for training a plate detector and, with less data, a character detector, but three things should be known before relying on it. The shipped split leaks, so any evaluation should regroup the images by source photo first; the cleaned table has the columns needed for that. The test split has almost no characters. And the collection is a set of pre-augmented crops from several countries, so results will not necessarily carry over to ordinary photos of Pakistani vehicles.

The limits of this analysis are just as real. A source image is defined by file name, which would merge two photos uploaded under the same name and split one photo uploaded under two. Only a handful of images were looked at, so the remarks about noise, cropping and country rest on a small sample and on naming patterns. Label correctness was not checked, and image properties such as brightness or blur were not measured. Useful next steps would be a split grouped by source, a visual audit of a few hundred labels, near-duplicate detection on the pixels to confirm the source grouping, and a comparison of detector scores on the shipped split against the grouped one to see how much the leak inflates them.